3.2 Row Operations with Determinants

The determinant of a triangular matrix is the product of its diagonal entries. Why?

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & 0 & \cdots & a_{nn} \end{vmatrix} \quad \text{----> expand in first column}$$

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} & \cdots & a_{2n} \\ 0 & a_{33} & a_{34} & \cdots & a_{3n} \\ 0 & 0 & a_{44} & \cdots & a_{4n} \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & 0 & \cdots & a_{nn} \end{vmatrix} \quad \text{----> first column again}$$

$$= a_{11} a_{22} \begin{vmatrix} a_{33} & a_{34} & a_{35} & \cdots & a_{3n} \\ 0 & a_{44} & a_{45} & \cdots & a_{4n} \\ 0 & 0 & a_{55} & \cdots & a_{5n} \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & 0 & \cdots & a_{nn} \end{vmatrix}$$

Continuing like this indeed becomes

$$a_{11} a_{22} \cdot \ldots \cdot a_{nn}$$

This also helps validate the fact that a triangular matrix is invertible exactly when every diagonal entry is nonzero.

So from this the motivation to convert a matrix to triangular (via row operations) is clear as it is very easy to find the determinant of a triangular matrix.

But how do row operations affect determinant? It was one thing when we were doing that to check invertibility as row operations do not affect that. But the exact value of the determinant is another story perhaps.

To help us figure this out it is useful to see how it works for 2 x 2 and think inductively and can also consider how cofactor expansion (usually in a row since those are what are being altered) works, i.e.

$$\det A = \sum_{j=1}^{n} a_{pj} C_{pj}$$

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(1) Scaling a row.

$$\begin{vmatrix} ra & rb \\ c & d \end{vmatrix} = ra(d) - rb(c) = r(ad - bc)$$

So scaling row in 2 x 2 matrix by r multiplies the determinant by that same r. So for 3 x 3 since expansion breaks down into a sum of 2 x 2, overall same for that as this shows:

$$\begin{vmatrix} \boxed{ra_{11} \quad ra_{12} \quad ra_{13}} \\ a_{21} \quad a_{22} \quad a_{23} \\ a_{31} \quad a_{32} \quad a_{33} \end{vmatrix}$$

$$= ra_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - ra_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + ra_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Just r times what it was prior, i.e. scales determinant by that same factor.

More generally, using the cofactor expansion in that same row that was scaled, each a_{pj} gets multiplied by r.

Meanwhile the cofactors are the same as the original matrix since after removing that scaled row, which happens in each of those cofactors, it just becomes the same matrix we obtained when doing the same cofactor expansion in the original matrix.

So since each a_{pj} is multplied by r, rest unaffected, overall determinant multplied by r.

So for any square matrix, scaling a row by a factor scales the determinant by the same factor.

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(2) Interchange two rows.

$$\begin{vmatrix} c & d \\ a & b \end{vmatrix} = bc - ad$$

Flips sign of determinant for 2 x 2. So for 3 x 3 will seemingly flip each cofactor and thereby flip sign of overall determinant.

But more generally in any square matrix if we interchange two consecutive rows and then expand in one of those two rows, all determinants in expansion stay the same as do the entries. The only change is in the $(-1)^{i+j}$ part of the cofactor as that will flip for each entry since each column number stays the same while the row number changes by one. Thus each cofactor switches sign.

Hence each term in summation switches sign so overall flips sign too.

So switching two consecutive rows always flips sign of determinant.

What if we switch any two rows, say rows j and k for some $j < k$?

Then applying a sequence of consecutive of row switches (flipping sign of determinant each time) can bring this about.

Switch the j-th row with the one underneath it repeatedly until it is right above row k, requires $k - j - 1$ such switches. Then moving row k up similarly until it is in the j-th position requires $k - j$ switches.

Thus we must switch consecutive rows $2k - 2j - 1$ times to bring this about. And since this is an odd number, no matter what j and k are, this flips overall sign of determinant.

Hence interchanging two rows always flips the sign of determinant for any size square matrix.

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(3) Replace row by itself plus multiple of another row.

See how this works in 3 x 3:

$$\begin{vmatrix} \boxed{a_{11} + ra_{21} \quad a_{12} + ra_{22} \quad a_{13} + ra_{23}} \\ a_{21} \quad a_{22} \quad a_{23} \\ a_{31} \quad a_{32} \quad a_{33} \end{vmatrix}$$

where we replaced first row by itself plus r times the second. Expanding in the first row we see that this is equivalent to:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} ra_{21} & ra_{22} & ra_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

which you can write out and see, largely because the cofactors are the same for the corresponding entries in the first row of the two matrices which makes this separation valid.

In this breakdown, the first matrix is just A and the second has first row a constant multiple of the second row. Thus the RREF of the second will have a zero row via replacing the first row by itself minus r times the second. Hence the second determinant is 0.

Thus the determinant is just overall $\det A$, and this works for any size matrix and any manipulation of this type (doesn't have to be the first two rows).

Therefore replacing a row by itself plus a multiple of another row doesn't change the determinant!

Remember that replacing say $R1$ by $2R1 + R2$ is two row operations not one! And the first of those, scaling the first row, does change determinant. If you think that is just one row operation, you will think it doesn't change determinant, and that will be wrong!

This is quite helpful since this third row operation is definitely the most used operation when trying to turn a square matrix into triangular.

And the fact that it doesn't alter the determinant makes our bookkeeping task much easier.

Do not think the idea used to prove adding multiple of of one row to another keeps determinant the same implies that det (A + B) is equal to det A + det B, and that's definitely not true! For one in that proof, it wasn't even sum, only first row was sum, the others would have created double original row if it was sum.

But either way, reason that worked was because it was only that one row. Trying to alter all rows like that at the same time, that idea definitely wouldn't work then.

It turns out that the operation that determinant behaves well under is product not sum, that is

$$\det(AB) = (\det A)(\det B)$$

whenever A and B are square matrices of same size.

Was mentioned that matrix multiplication is more important than addition, this is one reason. Even though multiplication of matrices is much more complicated than addition, multiplication works better in many ways.

This result is one, this also tells us that if A or B is not invertible, then AB is always not invertible as well, regardless of what the other of the two is. Definitely can't say that about $A + B$!

So why is determinant of product equal to product of determinants? Say E is an elementary matrix, then

$$\det(EA) = (\det E)(\det A)$$

Why? Well say E scales a row of I by r. Then $\det E = r$.

And from our knowledge of how elementary matrices work, EA will scale the same row in A by r, so $\det(EA)$ will be $r \cdot \det A$, i.e. $(\det E)(\det A)$, as desired.

Can check other two types of row operations similarly. Hence this works for one matrix being elementary.

Thus if A and B are both invertible write each as a product of elementary matrices (showed this is possible via the inverse algorithm), and applying the product idea successively will show for any two invertible matrices that determinant of product is product of determinants.

If one of A, B is not invertible, then $(\det A)(\det B)$ is certainly 0 and so is $\det(AB)$ so this still holds.

But why is latter 0 in that circumstance?

Say B is not invertible. Then by Invertible Matrix Theorem there is some nonzero y such that $By = 0$.

But then $(AB)y = 0$ as well so AB is also not invertible by that theorem, proving $\det(AB) = 0$.

If instead $\det A = 0$, again by Invertible Matrix Theorem, the columns of A do not span $\mathbb{R}^n$ so there exists some $z \in \mathbb{R}^n$ such that $Ax = z$ has no solution. But then $(AB)x = z$ certainly has no solution either (for then $A(Bx) = z$ would contradict what we just said prior). And that proves AB is not invertible, i.e. $\det(AB) = 0$.

This is a good illustration why the Invertible Matrix Theorem is quite helpful as it gave us several avenues to show this, and indeed we used different approaches depending on whether it was A or B which was not invertible.

Also have $\det(A^T) = \det A$

Why? Well note:

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc$$

so works for 2 x 2. Can then show 3 x 3 via connection with 2 x 2 via cofactors. Or generally just verify for any size matrix that expanding in the first row of A will be same calculation as expanding in first column of A^T Natural since that row in A is same as that column in A^T.

3.2.9

$$\begin{vmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 0 & 5 & 3 \\ 3 & -3 & -2 & 3 \end{vmatrix}$$

This is 4 x 4 with no row or column having more than a single 0 so using cofactors directly seems annoying. And since many entries are 1 or -1, making them 0 via row operations, we likely won't have to use fractions or scale rows to maintain integers. Fractions/scaling are annoying from a bookkeeping and calculation standpoint, and cofactors is annoying in calculation.

So thus a good situation on both counts to use row operations to simplify form before finding determinant.

Replace R3 with R3 + R1 and R4 with R4 - 3R1 (remember those are both type of row operation which doesn't alter determinant). That gives this equal to:

$$\begin{vmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & -1 & 2 & 3 \\ 0 & 0 & 7 & 3 \end{vmatrix}$$

Now replace R3 with R3 + R2 (again determinant stays the same):

$$= \begin{vmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 7 & 7 \\ 0 & 0 & 7 & 3 \end{vmatrix}$$

Finally replace R4 with R4 - R3 (once more this won't change the determinant).

$$= \begin{vmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 7 & 7 \\ 0 & 0 & 0 & -4 \end{vmatrix}$$

Now triangular so this equals

$$1(1)(7)(-4) = -28$$

Again this is almost certainly better than using cofactors since not that many 0s at start plus the row operations didn't involve fractions or scaling.

But again that won't always be the case, in some cases cofactors directly might be easier, you really have to use your judgement based on the matrix.

3.3 Cramer's Rule

Say we have a linear system with same number of equations as variables with unique solution (quite common with same number of equations as variables to have unique solution). Usually we solve that by row reducing augmented matrix. This provides another way of finding that solution.

As book says, this is not generally a "better" way, in fact it's usually longer if anything.

So why bother with it then? Well the ideas are practically interesting/useful. Comes up in situations you don't expect, e.g. the crossover with differential equations much later in the class.

Also shows a use for determinant beyond just using it to determine invertibility and gives us another way to find inverse of matrix.

For A an $n \times n$ matrix and b an $n \times 1$ matrix/vector, say we want to find (the assumed unique) solution to $Ax = b$.

Suppose that unique solution is denoted by:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

We'll solve for the x_i's one by one.

First some notation to help us.

<u>Notation:</u> In this setting for A being $n \times n$, b being $n \times 1$

$A_i(b)$ means the matrix where the i-th column of A is replaced by b (other columns of A stay same).

For this we have:

$$x_i = \frac{\det A_i(b)}{\det A}$$

Why? Consider the matrix:

$$A\,[I_i(x)]$$

Recall the second matrix in this product is just the identity matrix with x replacing the i-th column.

So what is this product ? Remember in any AB, the j-th column in AB is A times the j-th column in B. So here this is A times identity in every column except the i-th where it becomes Ax which equals b since we are assuming x is the solution to the stated system. Hence we have:

$$A\,[I_i(x)] = A_i(b)$$

Thus

$$\det\,(A\,[I_i(x)]) = \det\,[A_i(b)]$$

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$$(\det A)[\,\det I_i(x)\,] = \det\,[A_i(b)]$$

since determinant of product is product of determinants.
Then observe that

$$I_i(x) = \begin{bmatrix} 1 & 0 & \cdots & x_1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & x_2 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & \vdots & \vdots & \cdots & 0 \\ \vdots & \vdots & & x_i & 1 & \ddots & \vdots \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & x_n & 0 & \cdots & 1 \end{bmatrix}$$

That is, identity matrix, except i-th column is x instead of the usual e_i .

Expanding in i-th row, since that has all 0s except x_i that will give us x_i times the determinant of I_{n-1} as what remains there is just the identity matrix (just one size smaller). And since that is known to be 1, we have

$$\det\ [I_i(x)] = x_i$$

Thus the original equation becomes

$$(\det A)\, x_i = \det\,[A_i(b)]$$

So

$$x_i = \frac{\det\,[A_i(b)]}{\det A}$$

proving Cramer's rule.

Again whether this should be used in place of the usual methods is very unclear. After all we must do this for each i in $[1, n]$, so including A itself that is $n + 1$ determinants! So not short.

But ok these are all the matrix A, just maybe one one column modified. So not an independent calculation each time, but different enough at each step that it won't usually be short either.

Might not be so long if A and/or b has many 0 entries. But ok, realistically if that's case the other method (augmented matrix) would be proportionally easier as well so still not clear you'd want to do it this way.

Note also that $\det A = 0$ not allowed here as obviously calculation wouldn't work if it is. That guarantees A is invertible so solution is unique as it needs to be for Cramer's rule to work.

3.3.5

$$\begin{aligned} x_1 + x_2 &= 3 \\ -3x_1 + 2x_2 &= 0 \\ x_2 - 2x_3 &= 2 \end{aligned}$$

We have here:

$$A = \begin{bmatrix} \boxed{1 \quad 1 \quad 0} \\ -3 \quad 0 \quad 2 \\ 1 \quad 1 \quad 0 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$$

$$\det A = 1 \begin{vmatrix} 0 & 2 \\ 1 & -2 \end{vmatrix} - 1 \begin{vmatrix} -3 & 2 \\ 0 & -2 \end{vmatrix} = -2 - 6 = -8$$

Then by Cramer's Rule:

$$x_1 = \frac{\begin{vmatrix} 3 & 1 & 0 \\ \boxed{0 \quad 0 \quad 2} \\ 2 & 1 & -2 \end{vmatrix}}{-8} = \frac{-2 \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix}}{-8} = \frac{-2(3-2)}{-8} = \frac{1}{4}$$

$$x_2 = \frac{\begin{vmatrix} 1 & 3 & 0 \\ -3 & 0 & 2 \\ 0 & 2 & -2 \end{vmatrix}}{-8} = \frac{1\begin{vmatrix} 0 & 2 \\ 2 & -2 \end{vmatrix} + 3\begin{vmatrix} 3 & 0 \\ 2 & -2 \end{vmatrix}}{-8}$$

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$$x_2 = \frac{(0 - 4) + 3(-6)}{-8} = \frac{11}{4}$$

$$x_3 = \frac{\begin{vmatrix} 1 & 1 & 3 \\ -3 & 0 & 0 \\ 0 & 1 & 2 \end{vmatrix}}{-8} = \frac{3\begin{vmatrix} 1 & 3 \\ 1 & 2 \end{vmatrix}}{-8} = \frac{3(2 - 3)}{-8} = \frac{3}{8}$$

So solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} 2 \\ 22 \\ 3 \end{bmatrix}$$

In theory you could use Cramer's rule to solve for a single x_i if for whatever reason you only wanted one specific value rather than all of them.

When solving via other means (row reduction), mostly you are always solving for all of them inherently so Cramer's rule might be easier in that specific circumstance.

Ok, not very clear in what circumstance you might be motivated to do that, but helpful to know it's an option if you ever wanted to do that.